

Final Project Proposal

1. Certificate Name

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2. Project Title and Description

Title: Customer Segmentation & Churn Prediction

Customer churn is one of the most expensive problems in subscription-based and service industries, since acquiring a new customer typically costs far more than retaining an existing one. This project builds a system that first segments customers into behavioral groups based on their usage patterns, contract type, and spending, and then uses those segments — combined with the customer's raw attributes — to predict whether a given customer is likely to churn. The primary beneficiaries are telecom or subscription-service providers, whose retention teams can use the model's output to prioritize which customers to target with retention offers, and to understand *which type* of customer is at risk rather than treating all customers as a single uniform group.

3. Problem Type

Primary declared type: Classification (predicting a binary category — churn / no churn).

Clustering (K-Means) is used as an unsupervised **feature-engineering step**: customers are first grouped into segments based on behavioral and account features, and the resulting segment ID is added as an input feature to the supervised classifier. This keeps a single, deployable prediction target (churn probability) while still incorporating unsupervised learning into the pipeline, as required.

4. Dataset

- **Source:** [Telco Customer Churn dataset \(IBM sample dataset, hosted on Kaggle\)](#)
- **Size:** 7,043 rows (well above the 1,000-row minimum)
- **Target column (supervised):** `Churn` — a binary label (`Yes` / `No`) indicating whether the customer left the service within the last month
- **Main features:**
 - `tenure` — number of months the customer has stayed with the company
 - `MonthlyCharges` / `TotalCharges` — billing amount, used as spending indicators
 - `Contract` — contract type (month-to-month, one year, two year)
 - `InternetService` , `PhoneService` , `MultipleLines` — subscribed services
 - `OnlineSecurity` , `TechSupport` , `StreamingTV` , `StreamingMovies` — add-on services
 - `PaymentMethod` , `PaperlessBilling` — billing behavior
 - `gender` , `SeniorCitizen` , `Partner` , `Dependents` — demographic attributes

5. Algorithms to Train

#	Algorithm	Type	Why it fits
1	K-Means	Unsupervised (bootcamp)	Groups customers into behavioral segments (e.g., high-spend/long-tenure vs. low-spend/short-tenure) purely from feature similarity, with no label needed.
2	Logistic Regression	Supervised (bootcamp)	Provides an interpretable linear baseline, showing which features increase or decrease churn odds — important for business explainability.
3	Random Forest	Supervised (bootcamp)	Captures non-linear interactions between features (e.g., contract type combined with tenure) that logistic regression cannot, and is robust to outliers.
4	XGBoost	Supervised (self-researched)	Gradient boosting typically outperforms bagged trees on structured/tabular churn data and handles class imbalance well via <code>scale_pos_weight</code> tuning.

All four algorithms are distinct, and two (K-Means, Random Forest) plus Logistic Regression come directly from the bootcamp curriculum, satisfying the minimum requirement.

6. Evaluation Plan

- **Metrics used to compare all models:** Accuracy, Precision, Recall, and F1-score.
- **Metric used to select the best model: F1-score.**

Why F1: The churn dataset is class-imbalanced (churners are a minority class relative to retained customers). Accuracy alone can be misleading in this scenario, since a model that always predicts "no churn" would still score high on accuracy while being useless in practice. F1-score balances Precision (avoiding wasted retention spend on customers who wouldn't have churned anyway) and Recall (avoiding missed at-risk customers), making it the most appropriate single metric for this business problem.

7. Deployment Sketch

- **Framework:** Flask (lightweight, sufficient for a single `/predict` endpoint, consistent with prior project experience).
- **`/predict` endpoint — accepts (JSON):** `json { "tenure": 12, "MonthlyCharges": 70.35, "TotalCharges": 845.50, "Contract": "Month-to-month", "InternetService": "Fiber optic", "PaymentMethod": "Electronic check", "OnlineSecurity": "No", "TechSupport": "No", "SeniorCitizen": 0 }`
- **`/predict` endpoint — returns (JSON):** `json { "churn_prediction": "Yes", "churn_probability": 0.78, "segment_id": 2 }`
Where `segment_id` is the K-Means cluster assigned to the customer, and `churn_probability` is the predicted probability from the selected best-performing classifier (chosen by F1-score during evaluation).

8. Repository Plan

```
churn-segmentation-project/
├── dataset/
│   └── telco_customer_churn.csv
├── src/
│   ├── preprocess.py      # cleaning, encoding, train/test split
│   ├── clustering.py      # K-Means segmentation logic
│   ├── train.py           # trains Logistic Regression, Random Forest, XGBoost
│   └── evaluate.py        # computes Accuracy, Precision, Recall, F1 and selects best model
├── api/
│   └── app.py             # Flask app exposing /predict
├── models/
│   ├── kmeans_model.pkl
│   └── best_classifier.pkl
├── README.md
└── project_paper.md
```